and are called **semilunar valves**. Each semilunar valve is made of three half-moon shaped cusps. The myocardium of the ventricle is thrown into irregular muscular ridges called **trabeculae corneae**. The trabeculae corneae are modified into **chordae tendinae**. The opening and closing of the auriculo ventricular valves are achieved by the chordae tendinae. The chordae tendinae are attached to the lower end of the heart by papillary muscles. Heart receives deoxygenated blood from various parts of the body through the inferior venacava and superior venacava which open into the right auricle. Oxygenated blood from lungs is drained into the left auricle through four pulmonary veins.

7.4.1 Origin and Conduction of Heart Beat

The heart in human is myogenic (cardiomyocytes can produce spontaneous rhythmic depolarisation that initiates contractions). The sequence of electrical conduction of heart is shown in Figure 7.7. The cardiac cells with fastest rhythm are called the **Pacemaker cells**, since they determine the contraction rate of the entire heart. These cells are located in the right **sinuatrial (SA) node/ Pacemaker**. On the left side of the right atrium is a node called **auriculo ventricular node** (AV node). Two special cardiac muscle fibres originate from the auriculo ventricular node and are called the **bundle of His** which runs down into the interventricular septum and the fibres spread into the ventricles. These fibres are called the **Purkinje fibres**.

Pacemaker cells produce excitation through depolarisation of their cell membrane. Early depolarisation is slow and takes place by sodium influx and reduction in potassium efflux. Minimum potential is required to activate voltage gated calcium (Ca+) channels that causes rapid depolarisation which results in action potential. The pace maker cells repolarise slowly via K^+ efflux.

HEART BEAT- Rhythmic contraction and expansion of heart is called **heart beat**. The contraction of the heart is called **systole** and the relaxation of the heart is called **diastole**. The heart normally beats 70-72 times per min in a human adult. During each cardiac cycle two sounds are produced that can be heard through a **stethoscope**. The first heart sound lub is associated with

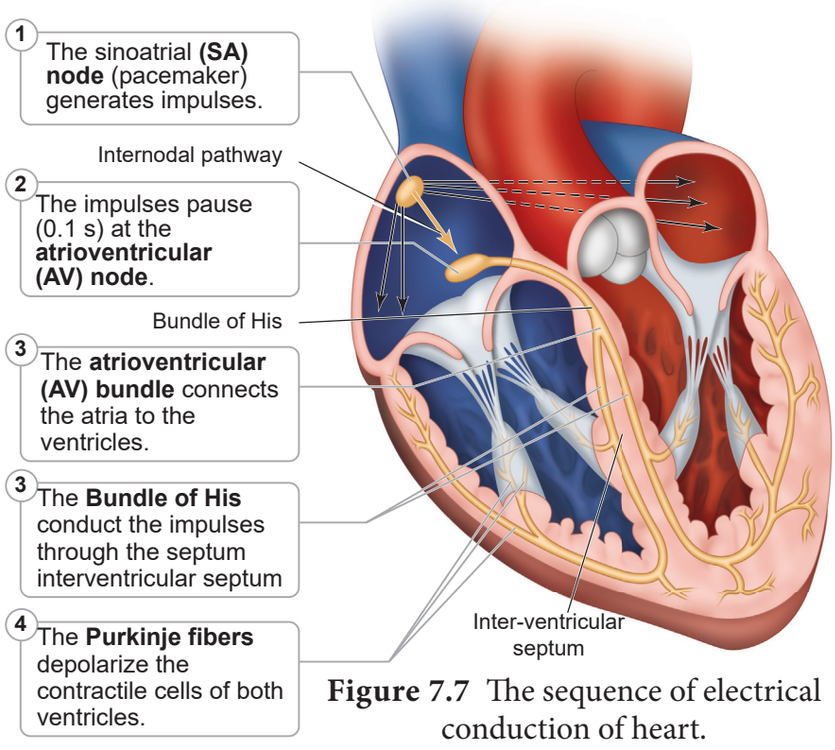


Figure 7.7 The sequence of electrical conduction of heart.

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